Law of sines

1 An oblique $\triangle ABC$ consists of angles A, B and C which appear opposite sides a, b and c respectively. State whether the following equations are true or false for this triangle:

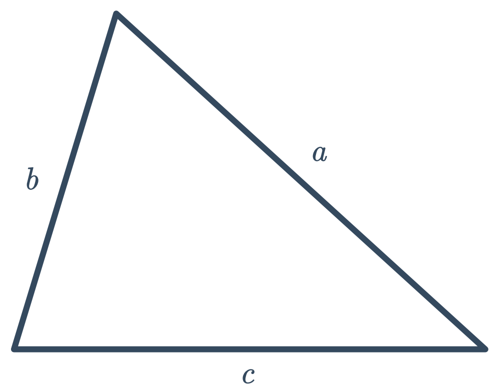
a $\dfrac{\sin B}{\sin C} = \dfrac{b}{c}$

b $\dfrac{a}{\sin A} = \dfrac{c}{\sin C}$

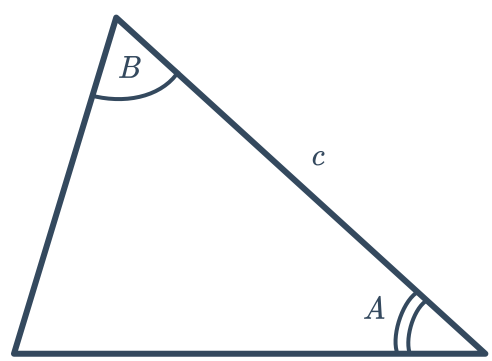
c $\dfrac{a}{\sin A} = \dfrac{\sin C}{c}$

d $\dfrac{\sin A}{a} = \dfrac{\sin B}{c}$

2 Consider a triangle where all three sides are known, but no angles are known.
Is there enough information to find all the angles in the triangle using only the law of sines? Explain your answer.



3 Consider a triangle where two of the angles and the side included between them are known.
Is there enough information to solve for the remaining sides and angle using the law of sines? Explain your answer.

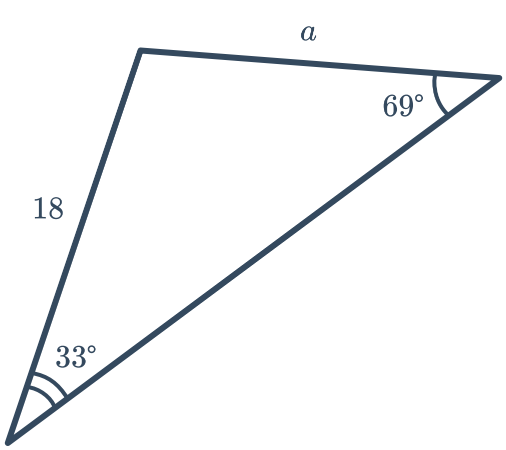


Unknown sides

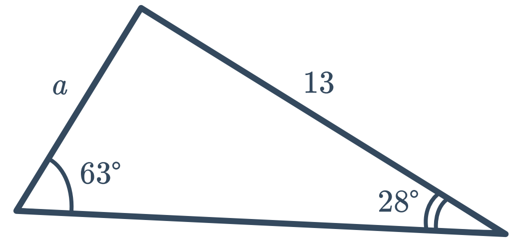
4 Consider the triangle $\triangle QUV$ where the side lengths q, u and v appear opposite $\angle Q, \angle U$ and $\angle V$.
If $q = 16$, $\sin V = 0.5$ and $\sin Q = 0.8$, find the length of side v.

5 Find the missing side length of the followin[redacted]ngles using the law of sines, rounding your answers to one decimal place:

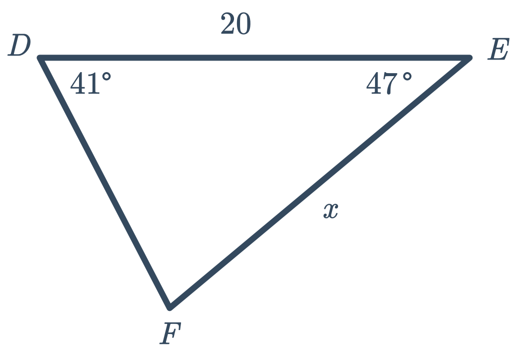
a



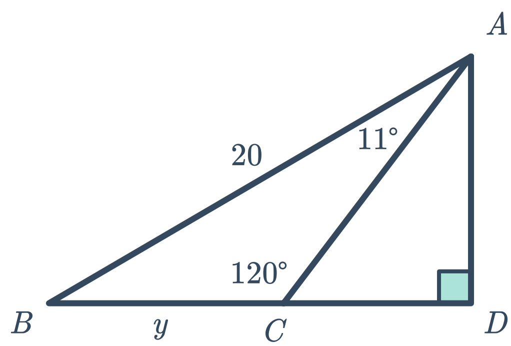
b



c



d

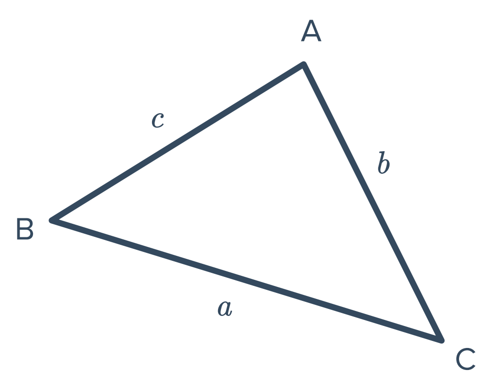


6 Consider the triangle with two interior angles $C = 72.53^\circ$ and $B = 31.69^\circ$, and one side length $a = 5.816 \text{ m}$:

a Find the measure of $\angle A$.

b Find the length of side b, rounded to three decimal places.

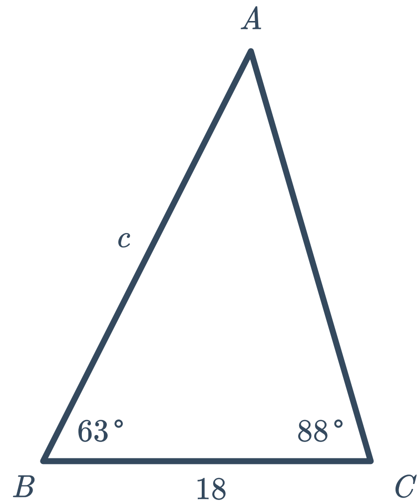
c Find the length of side c, rounded to three decimal places.



7 Consider the given triangle:

a Find the measure of $\angle BAC$.

b Find the length of side c.

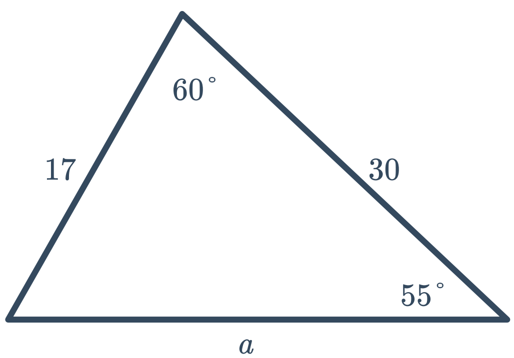


8 For each of the following:

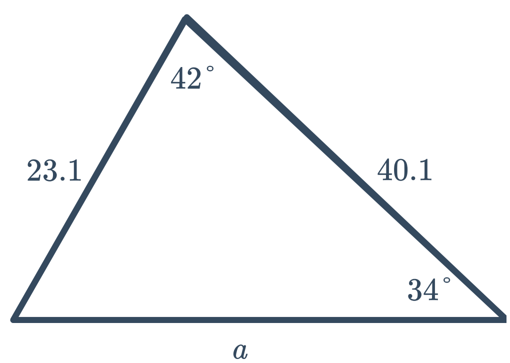
i State the range of possible lengths of the unknown side as an inequality.

ii Find the unknown side length, rounded to two decimal places.

a



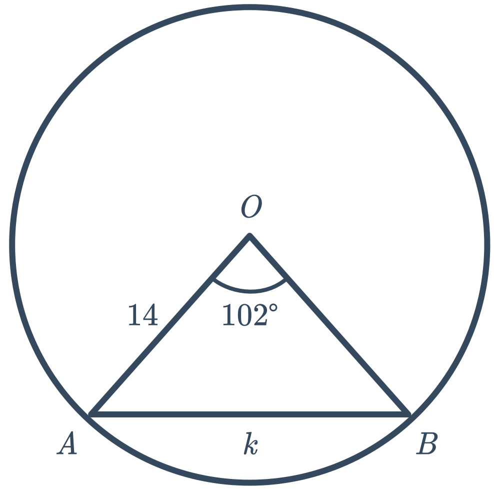
b



9 Consider the following diagram:

a Find the measure of $\angle OBA$.

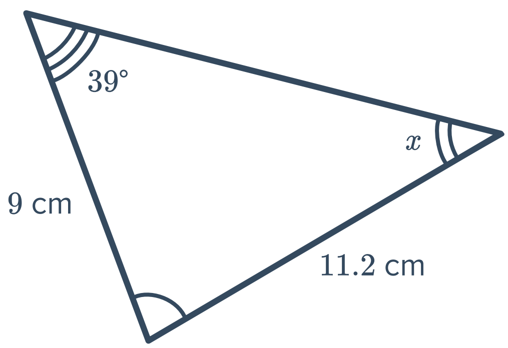
b Find the length of side k, rounded to two decimal places.



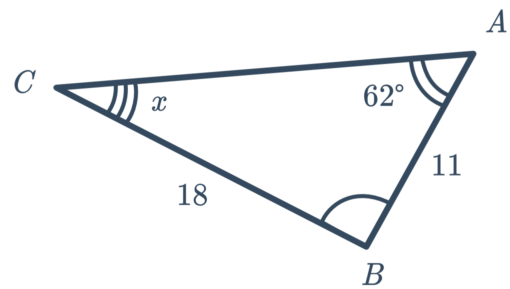
Unknown angles

10 Find the measure of angle x in the following triangles, rounded to one decimal place:

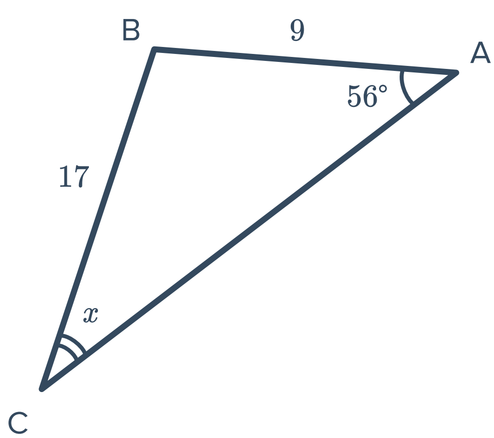
a



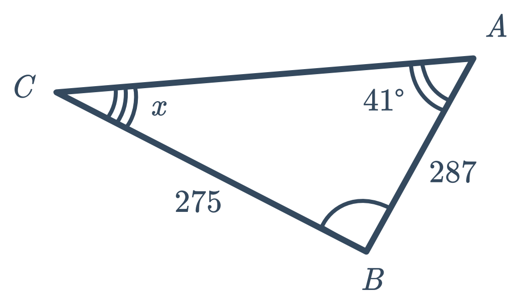
b



c

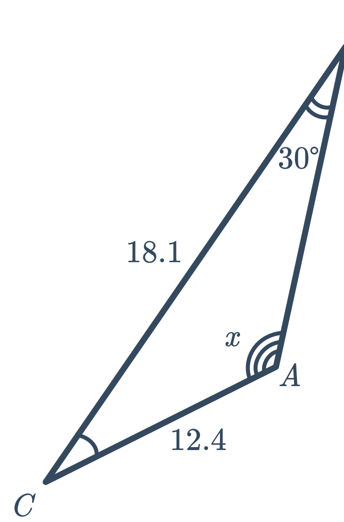


d

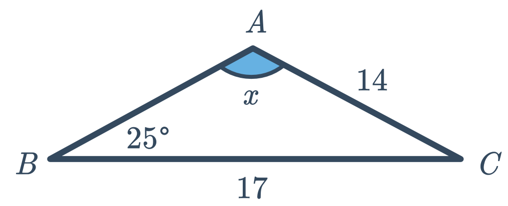


11 Find the measure of obtuse angle x in the following triangles, rounded to two decimal places:

a



b



Ambiguous case

12 Determine the number of possible triangles given the following measurements. Assume a is the side length opposite $\angle A$ and b is the side length opposite $\angle B$.

a $a = 50,\ b = 58$ and $A = 60^\circ$

b $a = 23,\ b = 21$ and $A = 35^\circ$

c $B = 30^\circ,\ b = 4$ and $c = 8$

d $a = 39,\ b = 32$ and $B = 50^\circ$

13 Determine whether the following sets of measurements determine a unique triangle, assuming a, b and c are the side lengths opposite $\angle A$, $\angle B$ and $\angle C$ respectively:

a $B = 40^\circ, b = 2, c = 5$

b $a = 6, b = 3, c = 27$

c $a = 3, b = 4, c = 5$

d $a = 80^\circ, b = 20^\circ, c = 80^\circ$

e $a = 5, b = 6, C = 80^\circ$

f $A = 50^\circ, B = 30^\circ, c = 8$

g $a = 20^\circ, b = 40^\circ, c = 120^\circ$

h $a = 5, b = 12, c = 13$

14 $\triangle ABC$ consists of angles A, B and C which appear opposite sides a, b and c respectively. Given the following, state whether solving $\triangle ABC$ results in the ambiguous case:

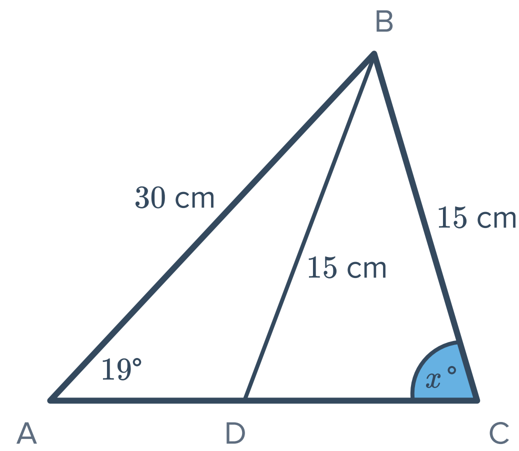
a If a, b and c are known.

b If $\angle A$, $\angle B$ and a are known.

c If $\angle A$, a and c are known.

d If a, $\angle B$ and c are known.

15 Consider $\triangle ABC$ below:

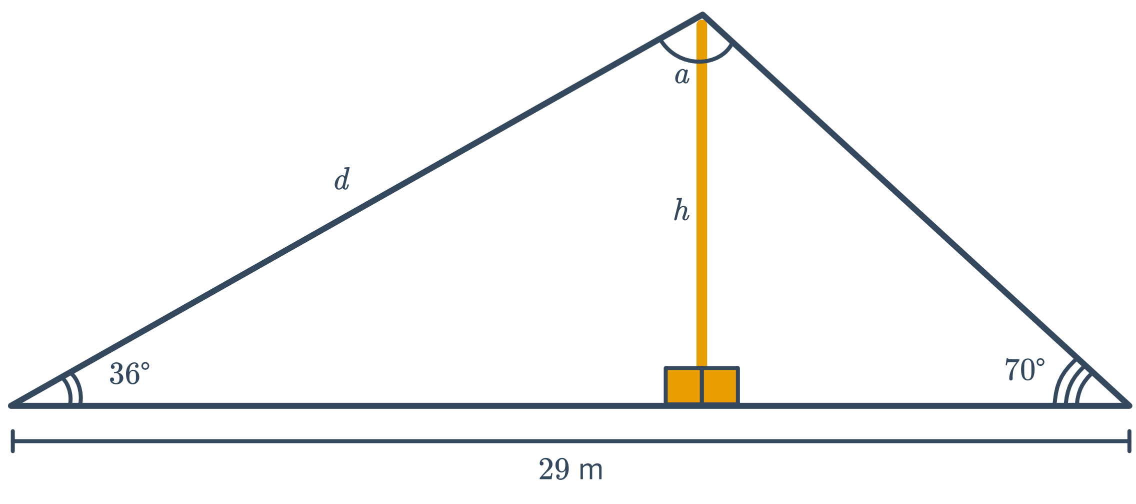


a. Find the measure of angle x, given that x is acute. Round your answer to the nearest degree.

b. Find the measure of $\angle ADB$ to the nearest degree, given that $\angle ADB > x$.

Applications

16 Two wires help support a tall pole. One wire forms an angle of 36° with the ground and the other wire forms an angle of 70° with the ground. The wires are 29 m apart.

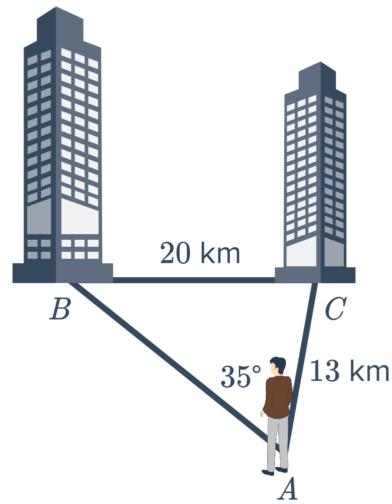


a. Find the measure of angle a, the angle made between the two wires at the top of the pole.

b. Find d, the length of longest wire in meters. Round your answer to two decimal places.

c. Calculate h, the height of the pole in meters. Round your answer to two decimal places.

17 Dave is standing on a hill and can see two buildings in the distance. Suppose the buildings are $20\ \mathrm{km}$ apart. Dave is $13\ \mathrm{km}$ from one building and the angle between the two lines of sight to the buildings is 35°.

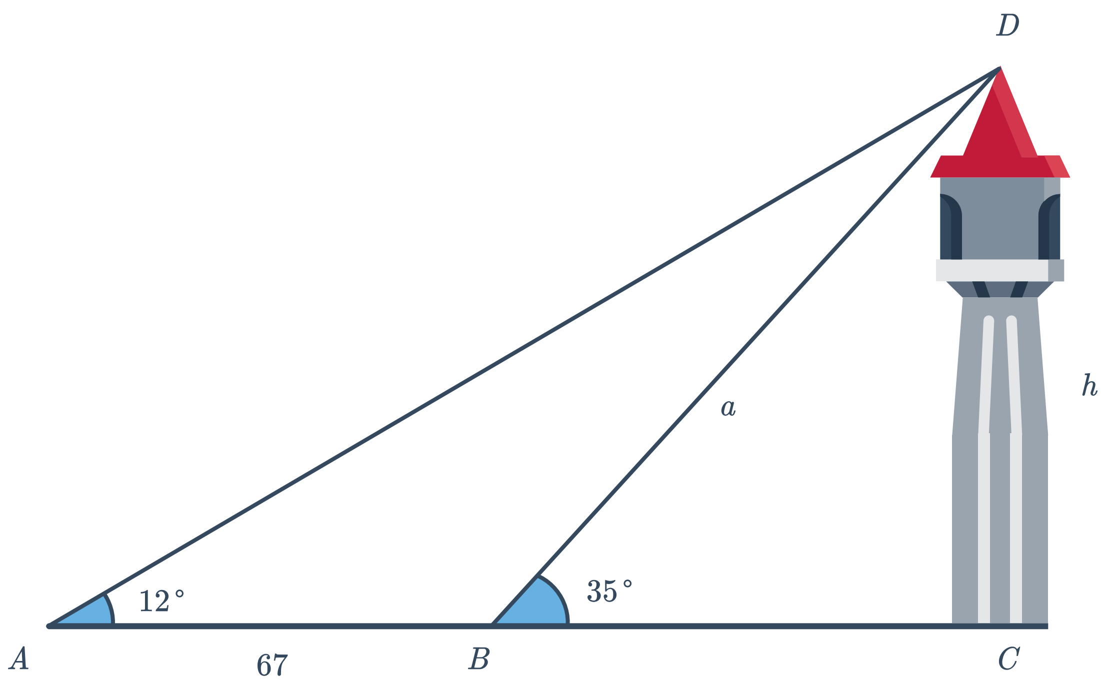


a Find the measure of $\angle ABC$. Round your answer to two decimal places.

b Find the measure of $\angle BCA$. Round your answer to two decimal places.

c Find the distance from Dave to the building at B. Round your answer to one decimal place.

18 Mae observes a tower at an angle of elevation of 12°. The tower is perpendicular to the ground. Walking $67\ \mathrm{yd}$ towards the tower, she finds that the angle of elevation increases to 35°.



a Find the measure of $\angle ADB$.

b Find the length of the side a. Round your answer to two decimal places.

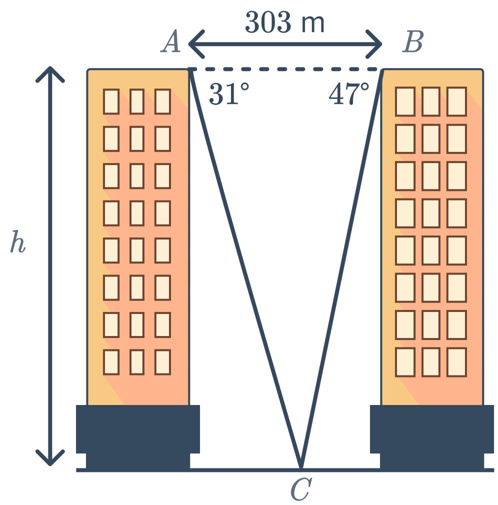
c Now, find the height h, of the tower. Round your answer to one decimal place.

19 To calculate the height of each apartment building, a surveyor measures the angles of depression from points A and B to C. From A, the measure of the angle of depression is 31° and from B the measure of the angle of depression is 47°.

a Find the measure of $\angle ACB$.

b If the distance between A and C is b m, find the value of b. Round your answer to two decimal places.

c If the buildings are h m tall, find the value of h. Round your answer to the nearest meter.



20 During soccer training, the coach marks out the perimeter of a triangular course that players need to run around. The diagram shows some measurements taken of the course, where side length $a = 14$ m:

a Find the measure of $\angle A$.

b Find the length of side c. Round your answer to two decimal places.

c Find the length of side b. Round your answer to two decimal places.

d Each player must sprint one lap around the triangle and then jog one lap around the triangle. This process is to be done 3 times by each player.
If Tara can run 280 m/min, and can jog at half the speed she runs, how long will this exercise take her? Round your answer to one decimal place.